

ENGR 102 Summer Bridge

Day 11 Student Quiz

While Loops, Update Order, Sentinels, and Termination

20 points • approximately 20-25 minutes

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Boolean-operator convention. Python operators appear as `and`, `or`, and `not`. Their lowercase spelling is required in code.

Show intermediate state for trace credit. Program credit requires one coherent solution. Unless a prompt says otherwise, give exact Python 3 output.

Question 1. Trace update order and termination.

5 points

```
value = 3
passes = 0
while value <= 20 and passes < 4:
    value = value * 2 - 1
    passes = passes + 1
print(value, passes)
```

Show every after-pass state and the first false condition; then give exact output.

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Question 2. Repair the update placement.

3 points

```
number = int(input())
while number != 0:
    if number > 0:
        print(number)
# intended: read the next number each pass
```

Add the one necessary line in the correct location and explain why placing it only inside the if is unsafe.

Question 3. Program construction

12 points

Write one complete Python program that satisfies every requirement.

- Read integer amounts until 0 is entered.
- Ignore negative amounts without adding or counting them.
- For positive amounts, compute count and total.
- After 0, print total and count, then print average when count is positive or empty when count is zero.
- Include one immediate-sentinel test.

[illegible]

Test evidence / expected result